// KEERTHAN TUNKOJU
*Entry-Level Software Developer | AI & ML Graduate*
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## PROFESSIONAL SUMMARY

AI/ML-focused Computer Science graduate and practicing GIS Engineer with experience in spatial data digitization, topology validation, and urban infrastructure mapping. Delivered end-to-end projects spanning convolutional neural networks for medical imaging and ML-driven agricultural prediction models. Currently contributing to large-scale urban GIS initiatives at NeoGeoInfo Technologies using ArcGIS, QGIS, and Google Earth.

## TECHNICAL SKILLS

**Languages:** Python, C, C++, JavaScript, SQL
**Web Technologies:** HTML5, CSS3, JavaScript
**ML / AI:** Convolutional Neural Networks (CNN), Supervised Learning, scikit-learn, Deep Learning
**GIS Tools:** ArcGIS, QGIS, Google Earth, Global Mapper
**GIS Skills:** Spatial Data Digitization, Topology Validation, Attribute QA/QC, Layer Management
**Tools & Version Control:** Git, GitHub, Microsoft Excel
**Databases:** SQL (MySQL)
**Soft Skills:** Problem Solving, Team Leadership, Communication

## PROFESSIONAL EXPERIENCE

**GIS Engineer** | *NeoGeoInfo Technologies Ltd.* — Feb 2026 – Present

- Perform utilities digitization of roads, footpaths, electric networks, drainage, water bodies, parks, and government properties for urban infrastructure projects.
- Execute spatial data validation and topology correction to ensure geometric integrity and compliance with project standards.
- Conduct attribute QA/QC using Excel, verifying data completeness, consistency, and accuracy across GIS layers.
- Create and manage GIS layers for multi-stakeholder urban projects, ensuring high-accuracy data delivery on time.
- Digitized urban infrastructure layers including roads, footpaths, electric network, drainage, water bodies, parks, buildings, and land parcels for the Greater Hyderabad Municipal Corporation (GHMC).
- Performed topology validation, attribute accuracy checks, and basemap correction using ArcGIS, QGIS, Google Earth, and Global Mapper.
- Delivered an accurate, production-ready urban GIS database supporting city planning and infrastructure management.

## INTERNSHIP EXPERIENCE

**Python Programming Intern** | *CodSoft* — Oct 2023 – Nov 2023

- Built a feature-complete To-Do List application in Python with task add, delete, and persistence functionality.
- Developed a GUI-based Calculator handling arithmetic, floating-point, and error edge cases.
- Implemented a Password Generator with configurable length, character sets, and entropy-based strength scoring.

**Web Developer Intern** | *Bharat Intern* — Sep 2023 – Oct 2023

- Designed and deployed a personal Portfolio website using HTML, CSS, and JavaScript showcasing projects and skills.
- Built a Temperature Converter tool supporting Celsius, Fahrenheit, and Kelvin conversions with real-time UI feedback.
- Recreated the Netflix homepage as a responsive UI exercise, demonstrating CSS layout and component structuring.

**Web Developer Intern** | *Intrainz* Jun 2023 – Aug 2023

- Developed a full Hotel website frontend using HTML, CSS, and JavaScript including booking UI, gallery, and contact form.
- Completed structured industrial training covering modern front-end development workflows and best practices.

## ACADEMIC PROJECTS

**Crop Yield Prediction Using Machine Learning**

- Designed and implemented an ML pipeline to predict crop yields based on soil type, weather conditions, crop management practices, and historical records, targeting a key challenge for India's agriculture-dependent economy.
- Applied and benchmarked multiple supervised learning techniques to model non-linear relationships between input features and crop yield output variables.
- Addressed key limitations of neural network approaches including relative error reduction and prediction efficiency, contributing to improved model robustness for real-world agricultural deployment.
- Synthesized a systematic review of AI-based Crop Yield Prediction (CYP) methods, extracting and evaluating feature sets used across the literature to inform model design.

**Deep Learning Methods for Identifying Brain Tumors**

- Built deep learning models using Convolutional Neural Networks (CNN) and watershed algorithms to perform automated semantic segmentation of brain MRI images, reducing the manual burden of clinical image analysis.
- Trained and evaluated CNN architectures on the BraTS dataset, experimenting with regularization techniques and hyperparameter tuning to optimize segmentation accuracy against heterogeneous tumor structures.
- Deployed a web application to expose the trained models to medical practitioners, enabling point-of-care inference without requiring local ML environment setup.
- Addressed challenges of spatial and structural heterogeneity in brain tumors to develop a dependable automated segmentation pipeline suitable for clinical settings.

## EDUCATION

**B.Tech in Computer Science & Engineering (AI & ML)** | *Sri Indu Institute of Engineering & Technology* 2020 – 2024

- CGPA: 6.6 / 10.0
- Relevant coursework: Machine Learning, Deep Learning, Data Structures & Algorithms, Database Management Systems, Computer Networks.

**Intermediate — MPC (Mathematics, Physics, Chemistry)** | *Narayana Jr College* 2018 – 2020

- Percentage: 75%

## CERTIFICATIONS

- Web Development Internship Certification — Intrainz (2023)
- Python Programming Internship Certification — CodSoft (2023)